Utkarsh Srivastava

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EDUCATION

Vellore Institute of Technology Aug. 2023 – Present
Bachelor of Technology in Computer Science and Engineering (Bioinformatics) Vellore, TN

- **CGPA**: 9.1 / 10.0 **(Ranked 6th out of 150 in specialization cohort)**

PUBLICATIONS & PREPRINTS

- **Utkarsh Srivastava**, Ashish R. Hota. "On Non-Stationary Constrained Multi-Armed Bandits." *Under review at Transactions on Machine Learning Research (TMLR)*, 2026. [OpenReview]
- **Utkarsh Srivastava**, Avidha Halder, Selvi M. "An Adaptive Security Selection Framework for Resource-Constrained IoT Devices Using the LinUCB Algorithm." *Accepted to IEEE INDISCON*, 2026. [PDF]
- **Utkarsh Srivastava**, Avidha Halder, Kathiravan S. "Byzantine-Robust Multi-Agent LLM Consensus." *Submitted to ACL Rolling Review (ARR)*, 2026. [OpenReview]

RESEARCH EXPERIENCE

Byzantine-Robust Multi-Agent LLM Consensus [OpenReview] | *Distributed AI, LLM Security* May 2026 – Present

- Formulated game-theoretic threat models and Byzantine fault-tolerant consensus mechanisms to defend decentralized multi-agent LLM systems against adversarial and malicious insider behaviors.
- Evaluated consensus convergence rates and robustness trade-offs under varying fractions of Byzantine agents across multi-turn collaborative decision tasks.

On Non-Stationary Constrained Multi-Armed Bandits [OpenReview] | *Advised by Dr. Ashish R. Hota* Jul. 2025 – Apr. 2026

- Engineered adaptive variants of Constraint Lower Confidence Bound (SW-CON-LCB, D-CON-LCB) using sliding-window and exponential discounting for piecewise-stationary environments.
- Derived theoretical upper bounds for suboptimal and infeasible arm selections, proving regret bounds of $\mathcal{O}(T^{\frac{1+v}{2}})$ under abrupt distributional shifts with $\mathcal{O}(T^v)$ breakpoints.
- Validated theoretical guarantees via Monte Carlo simulations on antimicrobial resistance (AMR) datasets; first-author manuscript under review at TMLR.

Adaptive Security Selection for Resource-Constrained IoT [PDF] | *Contextual Bandits* Dec. 2025 – Apr. 2026

- Designed a contextual bandit model using LinUCB to dynamically optimize cryptographic security primitives under strict computational budgets; accepted for publication at IEEE INDISCON 2026.

ChassisNet: Stress-Induced Interactome Routing [GitHub] | *Bioinformatics, Systems Biology* Jun. 2026 – Present

- Engineered a modular Python pipeline integrating BioGRID and STRING interaction databases to quantify molecular interactome routing redistribution under cellular stress.
- Validated network flow redistribution metrics against phenotypic expression benchmarks, generating automated network visualization and summary analyses.

TECHNICAL SKILLS

Languages: Python, C, R, SQL, MATLAB, Java, LaTeX

Research Areas: Online Learning (Bandits, MDPs), Stochastic Optimization, Distributed Consensus, Systems Biology

Tools & Libraries: NumPy, SciPy, NetworkX, PyTorch, CalC, AlphaFold2, PyMOL, Git

LEADERSHIP & SERVICE

Culinary Club VIT Jan. 2025 – Present

Chairperson & Co-Secretary Vellore Institute of Technology, Vellore

- Revitalized a dormant campus organization, scaling active membership from 300 to 1,600+ students and expanding programming from 1 to 20+ sold-out events per semester.
- Direct end-to-end logistics, vendor negotiations, and cross-team coordination for flagship workshops and campus-wide festivals exceeding 100+ attendees per session.